

25PY101: Engineering Physics

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Summative preparation

CFET

Instructions

- Answer all questions.
- Show all the calculation steps clearly.
- Draw neat and properly labeled diagrams.
- Use SI units wherever necessary.

Questions

Q1. Numerical Problem

An n-type silicon resistor has a length $L = 150\ \mu\text{m}$, width $W = 7.5\ \mu\text{m}$, and thickness $T = 1\ \mu\text{m}$. A voltage of $2\ \text{V}$ is applied across the length of the resistor. The donor impurity concentration varies linearly through the thickness of the resistor with $N_D = 2 \times 10^{16}\ \text{cm}^{-3}$ at the top surface and $N_D = 2 \times 10^{15}\ \text{cm}^{-3}$ at the bottom surface. Assume an average carrier mobility of $\mu_n = 750\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$.

- (a) What is the electric field in the resistor?
- (b) Determine the average conductivity of the silicon.
- (c) Calculate the current in the resistor.
- (d) Determine the current density near the top surface and the current density near the bottom surface .

Q2. Graphical / Sketch-Based Question

Sketch and label the following Hall effect measurement:

1. A Hall effect device is fabricated with the following geometry: $d = 5 \times 10^{-3}\ \text{cm}$, $W = 5 \times 10^{-2}\ \text{cm}$, and $L = 0.50\ \text{cm}$.
2. The electrical parameters measured are $I_x = 0.50\ \text{mA}$, $V_x = 1.25\ \text{V}$, and $B_z = 650\ \text{G}$
3. The Hall field is $E_H = -16.5\ \text{mV cm}^{-1}$.

Draw the circuit diagram and label the quantities.

Answer

Aim:

Topic:

Given Quantities:

Working Formulas:

Steps: